# Nipun Paul

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LinkedIn | GitHub | Google Scholar | Website

## RESEARCH INTERESTS

My research goal is to make machine learning reliable where data is scarce: models that learn well from limited or simulated data, are tested honestly against simpler baselines, and turn complex scientific data into insights people can act on. This goal grew out of my published work on low-resource machine translation and now drives my ongoing research in photonic sensor modelling and eDNA-based biodiversity monitoring.

## EDUCATION

**Ahsanullah University of Science and Technology (AUST)**, Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering April 2018 – January 2023
CGPA: 3.381 / 4.00 (Last 5 semesters: 3.656)

**Thesis:** *Decentralized Election System* — designed a blockchain-based voting architecture ensuring transparency, immutability, and secure vote validation using cryptographic verification.

## RESEARCH EXPERIENCE

**Independent Researcher** – ongoing, self-directed projects 2026 – Present

*GAN-Augmented Neural Network Modelling of a New Open-Channel PCF–SPR Plasmonic Sensor*

- Aims to compare ANN and XGBoost, each with and without WGAN-GP augmentation, for predicting the performance of a new open-channel sensor.

*From eDNA to Insights: An AI-Driven Environmental Monitoring Platform for Biodiversity Analysis*

- Aims to develop an AI-driven platform that turns eDNA survey data into interpretable biodiversity insights for environmental monitoring.

**Researcher / First Author – Bengali–English Neural Machine Translation** 2022 – 2023
*Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh*

- Conducted a controlled comparison of four sequence-to-sequence architectures (LSTM, GRU, BiLSTM, BiGRU) for Bengali-to-English translation in a low-resource setting.
- Merged four parallel corpora into a unified dataset and trained all models under matched experimental conditions.
- Evaluated with BLEU and ROUGE; BiLSTM performed best, with BLEU-1 to BLEU-4 scores of 47.4, 35.8, 32.0, and 22.8.

## PUBLICATIONS

**Paul, N.**, Faruki, I., Pranto, M. I., et al. (2023). *Bengali-English Neural Machine Translation Using Deep Learning Techniques.* 2023 International Conference on Electrical, Computer and Communication Engineering (ECCE), IEEE. DOI: 10.1109/ECCE57851.2023.10101491

## PROFESSIONAL EXPERIENCE

**Senior Software Engineer – Level II**, Enosis Solutions, Dhaka, Bangladesh Feb 2026 – Present
*Joined as Software Engineer (Level I) in Feb 2023; promoted annually through SE Level II and SSE Level I to SSE Level II.*

- Manage a full-scale Gantt-style plan scheduling platform end to end, from architecture to production, with task dependencies, timeline visualisation, and resource planning.
- Own backend and frontend delivery across five other internal products, including a Project Management System and a Payment Approval System with concurrency-safe approvals.

- Design shared building blocks and architecture reused across products, including JWT authentication, RBAC, audit logging, and real-time notifications.
- Lead cross-product initiatives: messaging integrations, authentication, CI/CD, and releases.
- Improve reliability and quality across products, and lead the team's move to an agentic, LLM-assisted development workflow, reducing development cycle time.

## SELECTED PROJECTS

**Strider – GPS Activity Tracking Platform**
Cross-platform app (Expo, Express, PostgreSQL/PostGIS) with geospatial track storage, variable-speed activity replay, and a synchronised elevation profile derived from recorded sensor traces.

**MotherCare – Maternity Health Platform** | github.com/Nipun4338/mothercare
Full-stack ASP.NET MVC platform with transactional payment integration and administrative dashboards.

**Project Boi – Book Exchange Marketplace** | github.com/Nipun4338/boi
Marketplace with authentication, real-time chat, wishlists, and an admin control panel.

## TECHNICAL EXPERTISE

**Programming:** Python, JavaScript, TypeScript, C++, C#, SQL
**Deep Learning & NLP:** PyTorch, TensorFlow/Keras, Hugging Face (Transformers, Datasets)
**ML & Scientific Computing:** scikit-learn, XGBoost, NumPy, pandas, Matplotlib, Jupyter
**Frameworks:** ReactJS, NodeJS, NestJS, ExpressJS, ASP.NET MVC, Expo/React Native
**Databases:** PostgreSQL (incl. PostGIS), MySQL, SQL Server, Firebase
**Tools:** Docker, Git, Bitbucket, Jira, Confluence, LaTeX, VS Code

## AWARDS AND ACHIEVEMENTS

- CODEWARE 2019 – 8th Place (Team: INT_CONST_2_1)
- ICPC Dhaka Regional Preliminary 2019 – 371st Place (Team: AUST Sacred Coders)
- Solved 1,200+ problems on 15 online judges: Codeforces (567), UVa (184), Toph (158), AtCoder (113), LightOJ (50), CodeChef (44), LeetCode (40), and others; 128 rated contests (profile)
- Codeforces peak 1377 (Pupil); Toph 1735; competed in Google Code Jam and Meta Hacker Cup
- Government Scholarship (Talentpool), 2015; Government Scholarship (General), 2013

## ACADEMIC AND COMMUNITY SERVICE

- **AUST Innovation and Design Club** (2019–2022) – Member to Sub-Executive Committee; led digitisation of event management workflows.
- **AUST ACM ICPC Lab** (2019–2021) – Competitive programming training and mentoring.
- **Lifetime Member**, Bangladesh Red Crescent Society

## LANGUAGES

Bengali (Native) | English (IELTS Academic **7.0**; L 7.5, R 8.0, W 6.5, S 6.0)

## REFERENCES

**Md. Khairul Hasan** | Associate Professor, CSE, AUST | khairul271276@aust.edu

**Raqeebir Rab** | Assistant Professor, CSE, AUST | raqeebir.cse@aust.edu